RouteShare:
An Intelligent Carpool System

Presented By

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INTRODUCTION

Problem Definition

Urban transportation faces issues like congestion, long travel times, and high commuting costs, while traditional ride-sharing lacks optimized matching, real-time updates, and reliable verification.

The problem is to develop an AI-driven carpool system that delivers optimized matching, secure transactions, real-time tracking, and a smooth user experience for affordable, efficient, and reliable daily commuting.

Feasibility Study

Technical Feasibility:

•AI-based routing integrated using OpenRouteService and geospatial algorithms.

•Secure authentication implemented via JWT with encrypted data handling.

•Integrates Stripe and AI modules Mistral for smooth operations.

Economic Feasibility:

•Uses mostly open-source tools, reducing dependency on paid services.

•Runs on standard hardware (CPU/GPU), reducing additional costs.

Operational Feasibility:

•User-friendly interface with simple ride creation and booking options.

•Real-time tracking and secured payments increase reliability and convenience.

Need and Significance

•Modern commuters expects **reliable**,**affordable** and **time efficient** travel options.

•Existing ride-sharing platforms lack **AI-driven route optimization**, proper user verification, and real-time coordination..

•An intelligent carpool system enhances **efficiency, safety, and travel convenience** through smart matching and automation.

Objectives

To develop an **AI-based intelligent carpool system** for daily commuters.

To provide ride matches that are **optimized, reliable, and time-efficient**.

To integrate **real-time tracking and smart routing** for improved accuracy.

To offer a **secure, easy-to-use, and seamless interface** for both drivers and passengers.

To ensure the system **safety, transparency, and user trust** throughout the ride process.

LITERATURE SURVEY

Existing Theoretical & Methodological Contributions

1.Haversine Formula Implementation (Sinnott, 1984)

Core Contribution: Accurate distance calculation on spherical surfaces

Foundation: Great-circle distance calculation between GPS coordinates
Application: Precise pickup/drop proximity calculations in RouteShare
Impact: Accurate confidence scoring based on real distances

2.Multi-Criteria Decision Analysis (MCDA) (Keeney & Raiffa, 1976)

Core Contribution: Systematic approach to multi-factor decision making

Foundation: Weighted scoring for complex decision scenarios
Application: RouteShare's 5-factor confidence scoring algorithm
Impact: Transparent, explainable ride quality assessments

3.Traveling Salesman Problem (TSP) (Dantzig et al., 1954)

Core Contribution: Optimization of routes with multiple stops

Foundation: Combinatorial optimization for route planning
Application: RouteShare's multi-leg journey planning
Impact: Optimal connecting ride combinations

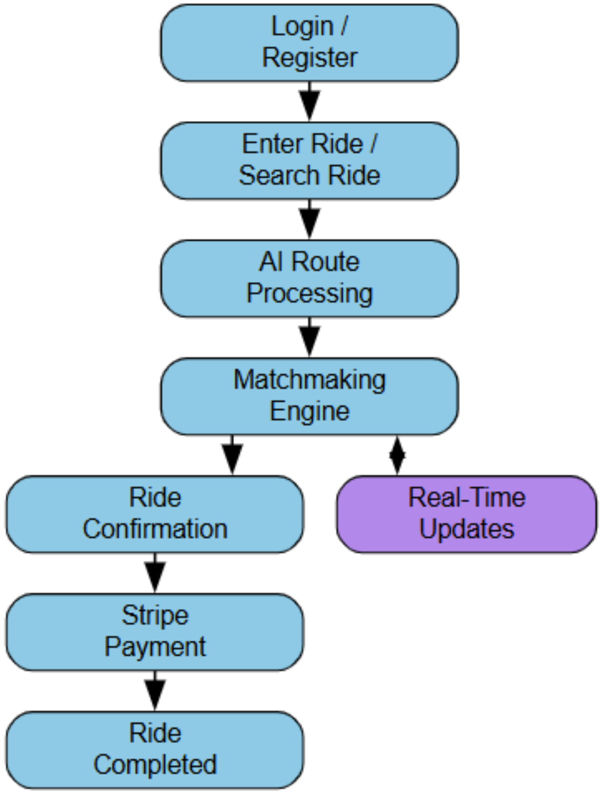
4.Dynamic Programming (Bellman, 1957)

Core Contribution: Breaking complex problems into simpler subproblems

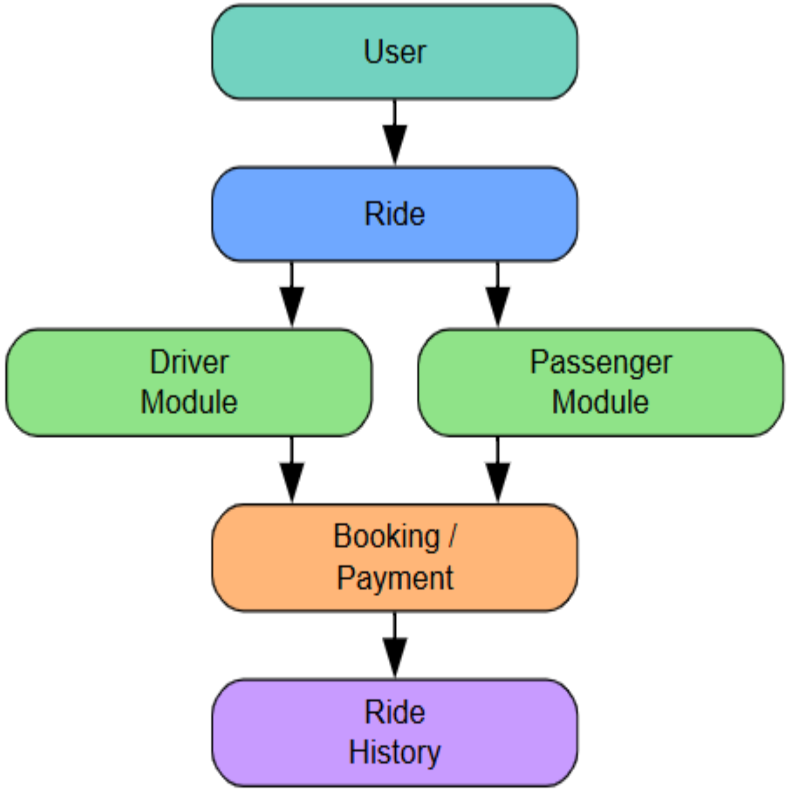
Foundation: Optimal substructure and overlapping subproblems
Application: RouteShare's multi-leg confidence calculation
Impact: Efficient computation of complex journey options

DESIGN ANALYSIS

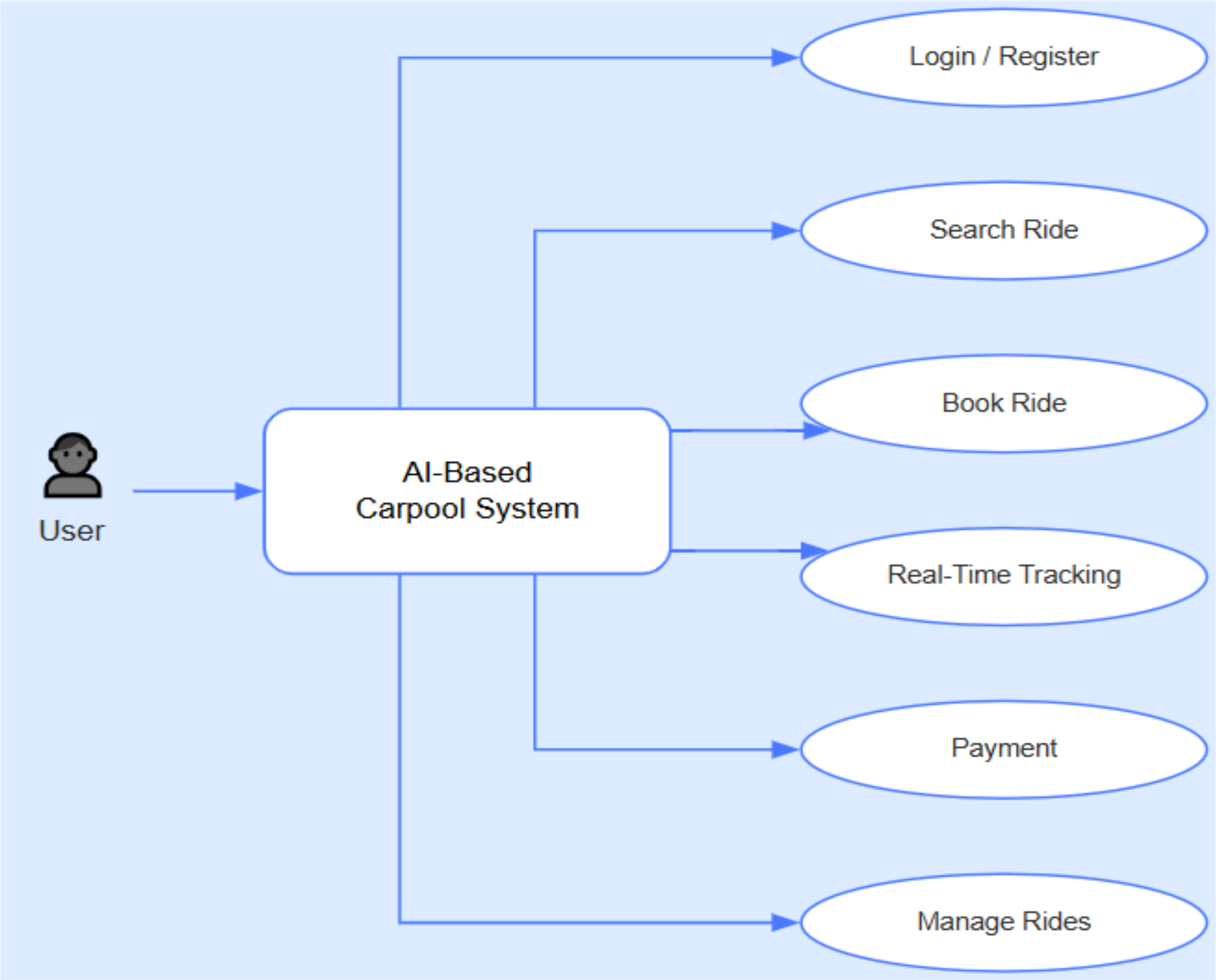
Flowchart



ER Diagram



Use Case Diagram



PROPOSED METHADOLOGY

Technical Architecture

User Service(Auth + Profiles)

Ride Service(Ride Management)

AI Service (Confidence Scoring + NLP Explanations)

Payment Service(Stripe)

PrimaryModel: "Hugging Face Mistral 8x7B"

The system uses a large AI language model from Hugging Face.It supports powerful reasoning and natural language processing.

CustomAI: "Custom heuristic algorithms"

Aside from the Mistral model, We have **custom-developed AI logic** for ride matching.

Used for:Confidence Scoring,Multi Leg Planning & Dynamic Pricing

Hardware and Software Requirements

Minimum Hardware Requirements

CPU: 4 cores
RAM: 8GB (16GB recommended)
Storage: 20GB free space
OS: Windows 10/11, macOS 10.15+, Ubuntu 18.04+

Software Requirements

Node.js: 18.0+ LTS
npm/yarn: Package manager
Git: Version control
VS Code: Recommended IDE
MongoDB: 6.0+ (local or Atlas)

CONCLUSION AND FUTURE WORK

Affiliated to GGSIP University, New Delhi
Approved by AICTE & Council of Architecture

RouteShare showcases how AI can improve ride-sharing through smart route optimization, transparent matching, and real-time decisions. It is scalable, user-friendly, and contributes to sustainable and efficient urban transportation while laying the foundation for future intelligent mobility solutions.

Future Work

- Personalization & Behavior Insights
- Blockchain & Web3 Integration

SCREENSHOTS

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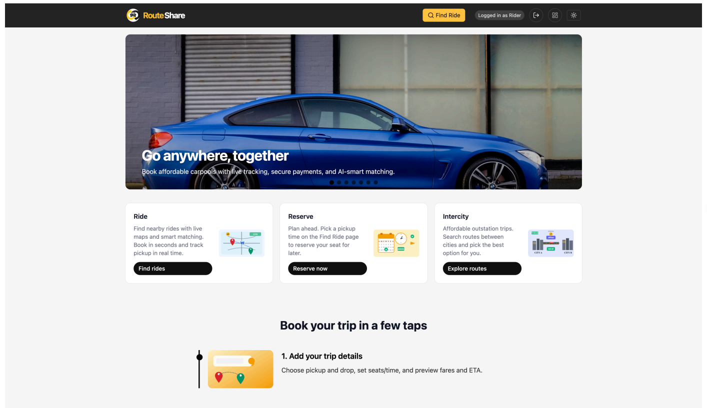

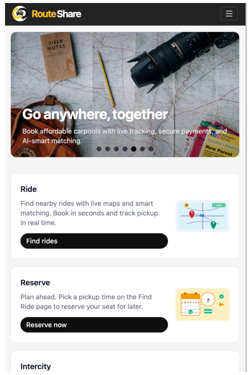

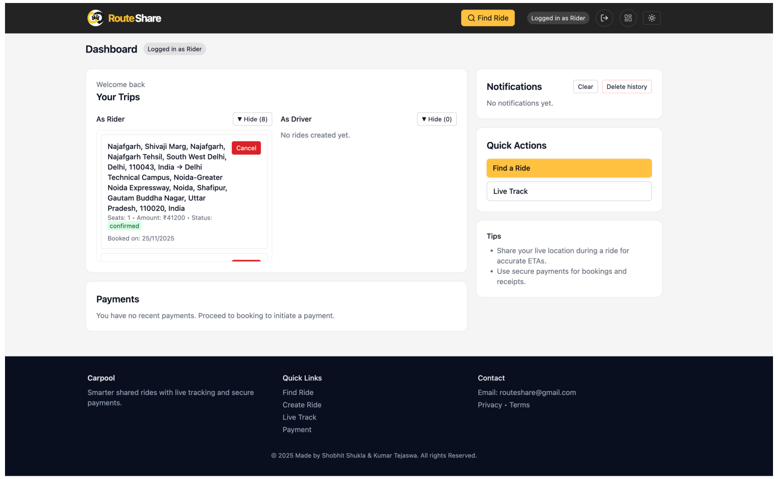

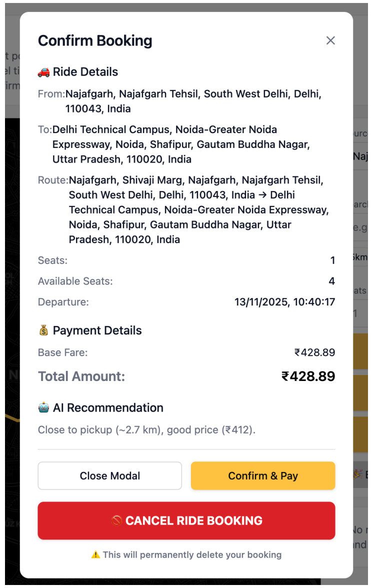

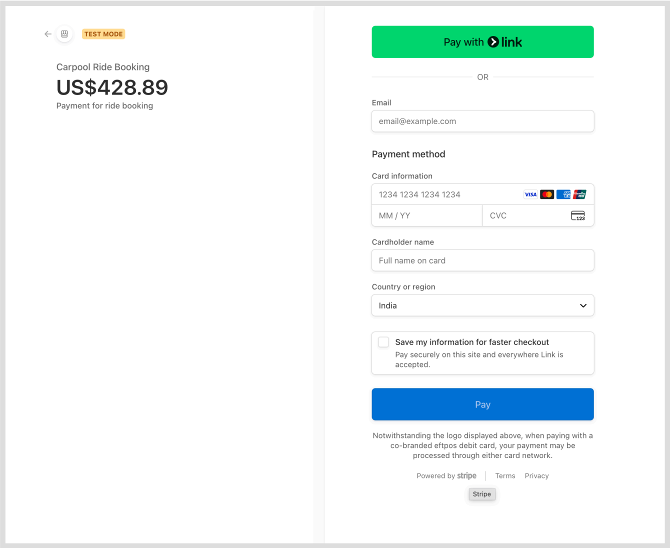

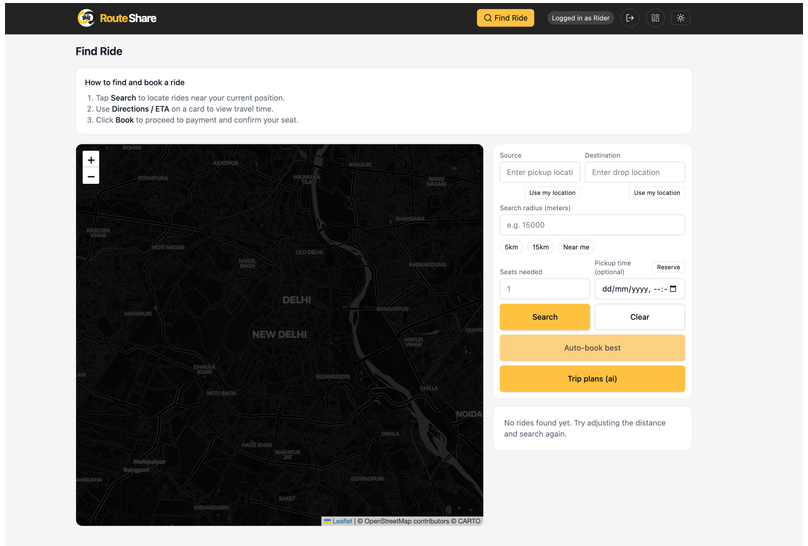

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